

Lecture 4

Further details of the Finite-Difference Method in 1D:

$\mathcal{O}(h^2)$ discretization of the Poisson equation with inhomogeneous coefficient function, approximation of boundary conditions, global error, singular perturbations, upwinding

Discretization of Poisson's equation with inhomogeneous coefficient (not in the book)

Laplace's operator with constant coefficient function

$$-k\Delta u = f$$

1D Poisson equation with homogeneous (constant) coefficient $k > 0$:

$$-ku'' = f, \quad x \in (0, 1)$$

FD approximation (uniform grid):

$$-ku_i'' = k \frac{-u_{i-1} + 2u_i - u_{i+1}}{h^2} + \mathcal{O}(h^2)$$

Laplace's operator with non-constant coefficient function

$$-\nabla \cdot (k \nabla u) = f$$

1D Poisson equation with inhomogeneous (nonuniform) coefficient $k(x) > 0$:

$$-(k(x)u')' = f, \quad x \in (0, 1)$$

How to obtain $\mathcal{O}(h^2)$ FD approximation in this case? Why not just take:

$$-(ku')'_i = k_i \frac{-u_{i-1} + 2u_i - u_{i+1}}{h^2} + \mathcal{O}(?)$$

Weighted second FD derivative

Previously we obtained $\mathcal{O}(h^2)$ FD approximation for u'' on a uniform grid as:

$$\begin{aligned}u_i'' &= \frac{u_{i+1} - 2u_i + u_{i-1}}{h^2} + \mathcal{O}(h^2) \\&= \frac{1}{h} \left(\frac{u_{i+1} - u_i}{h} - \frac{u_i - u_{i-1}}{h} \right) + \mathcal{O}(h^2)\end{aligned}$$

The idea is to use a more general, weighted formula:

$$(ku')'_i = \frac{1}{h} \left(a_{i+1} \frac{u_{i+1} - u_i}{h} - a_i \frac{u_i - u_{i-1}}{h} \right) + \mathcal{O}(h^2)$$

where $a_i = a(x_i)$ are the grid values of some function $a(x)$. We want to choose $a(x)$ in such a way that this gives an $\mathcal{O}(h^2)$ approximation of $(ku')'$ at x_i .

Using Taylor's expansions

Since,

$$\frac{u_{i+1} - u_i}{h} = u'_i + \frac{h}{2}u''_i + \frac{h^2}{6}u'''_i + \mathcal{O}(h^3)$$

$$\frac{u_i - u_{i-1}}{h} = u'_i - \frac{h}{2}u''_i + \frac{h^2}{6}u'''_i + \mathcal{O}(h^3)$$

we get

$$\begin{aligned} \frac{1}{h} \left(a_{i+1} \frac{u_{i+1} - u_i}{h} - a_i \frac{u_i - u_{i-1}}{h} \right) = \\ \frac{a_{i+1} - a_i}{h} u'_i + \frac{a_{i+1} + a_i}{2} u''_i + \frac{h(a_{i+1} - a_i)}{6} u'''_i + \mathcal{O}(h^2) \end{aligned}$$

Estimating the approximation error

Since, also,

$$(ku')' = ku'' + k'u', \quad (ku')'_i = k_i u''_i + k'_i u'_i$$

we get

$$\begin{aligned} & \frac{1}{h} \left(a_{i+1} \frac{u_{i+1} - u_i}{h} - a_i \frac{u_i - u_{i-1}}{h} \right) - (ku')'_i \\ &= \frac{a_{i+1} - a_i}{h} u'_i + \frac{a_{i+1} + a_i}{2} u''_i + \frac{h(a_{i+1} - a_i)}{6} u'''_i + \mathcal{O}(h^2) - k_i u''_i - k'_i u'_i \\ &= \left(\frac{a_{i+1} - a_i}{h} - k'_i \right) u'_i + \left(\frac{a_{i+1} + a_i}{2} - k_i \right) u''_i + \frac{h(a_{i+1} - a_i)}{6} u'''_i + \mathcal{O}(h^2) \end{aligned}$$

Conditions for second-order approximation

The error will be $\mathcal{O}(h^2)$ if

$$\begin{aligned}\frac{a_{i+1} - a_i}{h} &= k'_i + \mathcal{O}(h^2) \\ \frac{a_{i+1} + a_i}{2} &= k_i + \mathcal{O}(h^2)\end{aligned}$$

These are *sufficient conditions* for the approximation to be of the second-order. If these conditions are satisfied, we get:

$$\frac{1}{h} \left(a_{i+1} \frac{u_{i+1} - u_i}{h} - a_i \frac{u_i - u_{i-1}}{h} \right) - (ku')'_i = \mathcal{O}(h^2)$$

Possible choices of weights

Any of the following a_i 's will satisfy these conditions:

$$a_i = \frac{k(x_i) + k(x_{i-1})}{2} \quad \text{arithmetic mean of values}$$

$$a_i = k(x_i - 0.5h) \quad \text{value between } x_i \text{ and } x_{i-1}$$

$$a_i = \sqrt{k(x_i) k(x_{i-1})} \quad \text{geometric mean of values}$$

Note that choosing $a_i = k(x_i)$ will only give the first-order approximation.

Example: arithmetic mean

Constant coefficient function:

$$-ku_i'' = \frac{k}{h^2} (-u_{i-1} + 2u_i - u_{i+1}) + \mathcal{O}(h^2)$$

Varying coefficient function:

$$\begin{aligned} -(ku')'_i &= -\frac{1}{h} \left(\frac{k_{i+1} + k_i}{2} \frac{u_{i+1} - u_i}{h} - \frac{k_i + k_{i-1}}{2} \frac{u_i - u_{i-1}}{h} \right) + \mathcal{O}(h^2) \\ &= \frac{1}{h^2} \left(-\frac{k_i + k_{i-1}}{2} u_{i-1} + \frac{k_{i+1} + 2k_i + k_{i-1}}{2} u_i - \frac{k_{i+1} + k_i}{2} u_{i+1} \right) + \mathcal{O}(h^2) \end{aligned}$$

What is the system matrix? Is it still symmetric?

Approximation of boundary conditions

Steady-state temperature distribution

1D problem

$$-(ku')' = f, \quad x \in (0, L)$$

$$-k(0)u'(0) + \beta u(0) = \mu_1, \quad u(L) = \mu_2$$

describes the stationary distribution of temperature in a rod with one end ($x = L$) kept at temperature μ_2 , while the other end ($x = 0$) is exchanging heat with environment ($\beta \geq 0$).

$k(x) \geq c_1 > 0$ – thermal conductivity coefficient

$f(x)$ – density of thermal sources

BC's as additional equations

$N - 1$ algebraic equations at the inner points of the grid:

$$-\frac{1}{h} \left(a_{i+1} \frac{u_{i+1} - u_i}{h} - a_i \frac{u_i - u_{i-1}}{h} \right) = f_i, \quad i = 1, 2, \dots, N - 1$$

$$\frac{1}{h^2} (-a_i u_{i-1} + (a_i + a_{i+1}) u_i - a_{i+1} u_{i+1}) = f_i, \quad i = 1, 2, \dots, N - 1$$

Dirichlet BC:

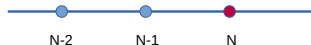
$$u_N = \mu_2$$

Robin BC:

$$-k(0)u'_0 + \beta u_0 = \mu_1,$$

Contains u'_0 , requires FD approximation.

Dirichlet BC



$$\frac{1}{h^2} (-a_{N-1}u_{N-2} + (a_{N-1} + a_N)u_{N-1} - a_N u_N) = f_{N-1}$$

$$u_N = \mu_2 \quad \text{Dirichlet BC}$$

Eliminating u_N :

$$\frac{1}{h^2} (-a_{N-1}u_{N-2} + (a_{N-1} + a_N)u_{N-1}) = f_{N-1} + \frac{a_N}{h^2}\mu_2$$

Robin BC 1



Using the forward one-sided FD approximation $u'_0 = (u_1 - u_0)/h + \mathcal{O}(h)$:

$$-k_0 \frac{1}{h} (u_1 - u_0) + \beta u_0 = \mu_1 \quad \text{Robin BC}$$

$$\frac{1}{h^2} (-a_1 u_0 + (a_1 + a_2) u_1 - a_2 u_2) = f_1$$

To recognize matrix elements, rewrite as:

$$\left(\beta + \frac{k_0}{h} \right) u_0 - \frac{k_0}{h} u_1 = \mu_1$$
$$\frac{1}{h^2} (-a_1 u_0 + (a_1 + a_2) u_1 - a_2 u_2) = f_1$$

Robin BC 2



Alternative: introduce an additional (virtual) grid point $x_0 - h = x_{-1}$ and use the central FD approximation $u'_0 = (u_1 - u_{-1})/(2h) + \mathcal{O}(h^2)$:

$$-k_0 \frac{1}{2h} (u_1 - u_{-1}) + \beta u_0 = \mu_1 \quad \text{Robin BC}$$

$$\frac{1}{h^2} (-a_0 u_{-1} + (a_0 + a_1)u_0 - a_1 u_1) = f_0 \quad \text{extra eq.}$$

$$\frac{1}{h^2} (-a_1 u_0 + (a_1 + a_2)u_1 - a_2 u_2) = f_1$$

Eliminate virtual unknown: $u_{-1} = -(2h/k_0)\beta u_0 + u_1 + (2h/k_0)\mu_1$:

$$\frac{1}{h^2} \left(\left(\frac{2h\beta}{k_0} a_0 + a_0 + a_1 \right) u_0 - (a_0 + a_1)u_1 \right) = f_0 + \frac{2a_0\mu_1}{hk_0}$$

The local error goes up to $\mathcal{O}(h)$ after elimination of u_{-1} .

Periodic BC's

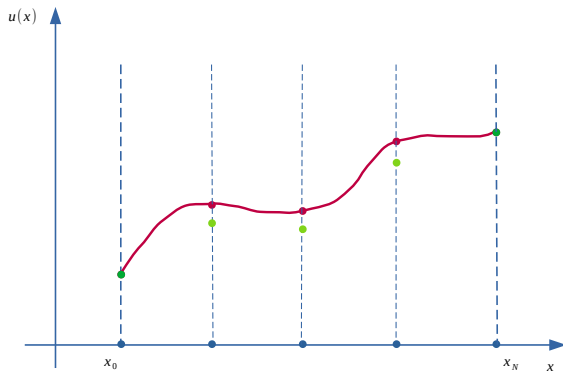
Boundary conditions that enforce periodicity of the solution outside the interval $[0, L]$, i.e. $u(x + L) = u(x)$, may be given in the form

$$u(0) = u(L)$$

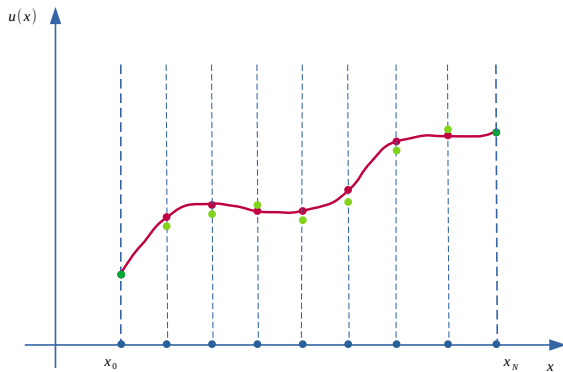
$$u'(0) = u'(L)$$

Global error of the numerical solution

Concept of error, grid-step h



Concept of error, grid-step $h/2$



Measure of solution error

In this discussion we need to distinguish between the exact and numerical solutions at x_i :

- Let $u_i = u(x_i)$ be the exact solution
- Let $\tilde{u}_i$ be the numerical solution at x_i

How to measure the error? There are many ways, but we shall consider the so-called Root Mean Squared Error (RMSE):

- Locally, at x_i the squared error is $|u_i - \tilde{u}_i|^2$
- Globally, at all x_i , $i = 1, \dots, N - 1$; the RMSE is:

$$\epsilon = \sqrt{\frac{1}{N-1} \sum_{i=1}^{N-1} |u_i - \tilde{u}_i|^2}$$

Relation to L_2 norm

If $u(x)$ and $v(x)$ are functions on $[a, b]$, and $u(a) = v(a)$, $u(b) = v(b)$, then the L_2 norm of their difference is:

$$\|u - v\|_2 = \sqrt{\int_a^b |u(x) - v(x)|^2 dx} \approx \sqrt{\sum_{i=1}^{N-1} |u_i - v_i|^2 \frac{b-a}{N-1}}$$

If $\mathbf{u}$ and $\mathbf{v}$ are vectors in $\mathbb{R}^{N-1}$, then the algebraic (Euclidean) L_2 norm, used in `np.linalg.norm()`, is:

$$\|\mathbf{u} - \mathbf{v}\|_2 = \sqrt{\sum_{i=1}^{N-1} |u_i - v_i|^2}$$

The RMSE, also called $L_{2,h}$ norm, is related to the algebraic L_2 norm and the L_2 norm for functions as follows:

$$\text{RMSE}(\mathbf{u}, \mathbf{v}) = \|\mathbf{u} - \mathbf{v}\|_{2,h} = \frac{1}{\sqrt{N-1}} \|\mathbf{u} - \mathbf{v}\|_2 \approx \frac{1}{\sqrt{b-a}} \|u - v\|_2$$

Exact discrete problem

We know that the local approximation error is:

$$-u'' = \frac{1}{h^2}(-u_{i-1} + 2u_i - u_{i+1}) + \mathcal{O}(h^2) = \frac{1}{h^2}(-u_{i-1} + 2u_i - u_{i+1}) + p_i h^2,$$

where p_i is some constant. Hence, with Dirichlet BC's, the exact solution u_i , $i = 1, \dots, N-1$; satisfies:

$$\begin{aligned}\frac{1}{h^2}(2u_1 - u_2) &= f_1 + u_0 h^{-2} + p_1 h^2 \\ \frac{1}{h^2}(-u_1 + 2u_2 - u_3) &= f_2 + p_2 h^2 \\ &\vdots \\ \frac{1}{h^2}(-u_{N-2} + 2u_{N-1}) &= f_{N-1} + u_N h^{-2} + p_{N-1} h^2\end{aligned}$$

where p_i , $i = 1, \dots, N-1$; are some unknown but finite constants.

Numerical problem

The numerical solution $\tilde{u}_i$, $i = 1, \dots, N - 1$; satisfies

$$\frac{1}{h^2}(2\tilde{u}_1 - \tilde{u}_2) = f_1 + u_0 h^{-2}$$

$$\frac{1}{h^2}(-\tilde{u}_1 + 2\tilde{u}_2 - \tilde{u}_3) = f_2$$

$$\vdots$$

$$\frac{1}{h^2}(-\tilde{u}_{N-2} + 2\tilde{u}_{N-1}) = f_{N-1} + u_N h^{-2}$$

Algebraic formulation

Introduce the vector of approximation errors:

$$\begin{bmatrix} p_1 h^2 \\ p_2 h^2 \\ \vdots \\ p_{N-1} h^2 \end{bmatrix} = h^2 \begin{bmatrix} p_1 \\ p_2 \\ \vdots \\ p_{N-1} \end{bmatrix} = h^2 \mathbf{p}$$

the vectors of exact and numerical solutions, and the source vector:

$$\mathbf{u} = \begin{bmatrix} u_1 \\ u_2 \\ \vdots \\ u_{N-1} \end{bmatrix}, \quad \tilde{\mathbf{u}} = \begin{bmatrix} \tilde{u}_1 \\ \tilde{u}_2 \\ \vdots \\ \tilde{u}_{N-1} \end{bmatrix}, \quad \mathbf{f} = \begin{bmatrix} f_1 + u_0 h^{-2} \\ f_2 \\ \vdots \\ f_{N-1} + u_N h^{-2} \end{bmatrix}$$

we arrive at two problems with the same negative Laplacian system matrix $A \in \mathbb{R}^{(N-1) \times (N-1)}$:

$$A\mathbf{u} = \mathbf{f} + h^2 \mathbf{p},$$

$$A\tilde{\mathbf{u}} = \mathbf{f}.$$

The error vector

Subtracting these two problems we get:

$$\begin{aligned}A\mathbf{e} &= h^2\mathbf{p}, \\ \mathbf{e} &= \mathbf{u} - \tilde{\mathbf{u}},\end{aligned}$$

where $\mathbf{e} \in \mathbb{R}^{N-1}$ is the vector of all errors with components $u_i - \tilde{u}_i$. We want to estimate the $L_{2,h}$ norm of this error vector.

Global solution error (Dirichlet BC's)

Theorem: The global numerical solution error of the Poisson equation $-u'' = f$ with Dirichlet BC's for $\mathcal{O}(h^2)$ finite-difference method is $\|\mathbf{e}\|_{2,h} = \mathcal{O}(h^2)$

Proof: The system matrix A obtained after FD discretization is symmetric positive definite. Hence, $A = Q\Lambda Q^T$ and $A^{-1} = Q\Lambda^{-1}Q^T$, with $\lambda_{\max}(A^{-1}) = 1/\lambda_{\min}(A)$. Using $\|A\|_2 = \sqrt{\lambda_{\max}(A^T A)}$, we obtain

$$A\mathbf{e} = h^2\mathbf{p}, \implies \mathbf{e} = h^2A^{-1}\mathbf{p}$$

$$\begin{aligned}\|\mathbf{e}\|_2 &= \|h^2A^{-1}\mathbf{p}\|_2 \leq h^2 \|A^{-1}\|_2 \|\mathbf{p}\|_2 = h^2 \sqrt{\lambda_{\max}(A^{-T}A^{-1})} \|\mathbf{p}\|_2 \\ &= \frac{h^2}{\lambda_{\min}(A)} \|\mathbf{p}\|_2 = h^2 \frac{(x_N - x_0)^2 + \mathcal{O}(h^2)}{\pi^2} \|\mathbf{p}\|_2\end{aligned}$$

Hence, $\|\mathbf{e}\|_{2,h} \leq h^2 \frac{(x_N - x_0)^2 + \mathcal{O}(h^2)}{\pi^2} \|\mathbf{p}\|_{2,h} = \mathcal{O}(h^2)$, since $\|\mathbf{p}\|_{2,h}$ is bounded.

Case of Dirichlet and Neumann boundary conditions

Continuous problem:

$$-u'' = f, \quad u(x_0) = 0, \quad u'(x_N) = 0$$

Discretized problem:

$$\mathcal{O}(h^2) : \quad -\frac{1}{h^2} (u_{i-1} - 2u_i + u_{i+1}) = f_i, \quad i = 1, \dots, N-1;$$

$$\mathcal{O}(h) : \quad \frac{1}{h} (u_N - u_{N-1}) = 0$$

The FD matrix $\tilde{A} \in \mathbb{R}^{N \times N}$ is still symmetric

$$\tilde{A} = \frac{1}{h^2} \begin{bmatrix} 2 & -1 & 0 & 0 & \dots & 0 \\ -1 & 2 & -1 & 0 & \dots & 0 \\ \cdot & & & & & \cdot \\ \cdot & & & & & -1 \\ 0 & 0 & 0 & \dots & -1 & 1 \end{bmatrix}$$

Error for $\mathcal{O}(h^2)$ at inner points and $\mathcal{O}(h)$ at boundaries

Since the FD approximation is $\mathcal{O}(h^2)$ at the inner points of the grid but is $\mathcal{O}(h)$ at the boundary (Neumann BC at x_N), we have the following exact discrete problem:

$$\tilde{A}\mathbf{u} = \mathbf{f} + h^2\mathbf{p} + h\mathbf{q}, \quad \text{and} \quad \|\mathbf{p}\|_{2,h}, \|\mathbf{q}\|_{2,h} \text{ remain bounded as } h \rightarrow 0$$

Moreover, $\mathbf{q} = [0, \dots, 0, q_N]^T$. The numerical solution $\tilde{\mathbf{u}}$ satisfies $\tilde{A}\tilde{\mathbf{u}} = \mathbf{f}$. The error vector $\mathbf{e} = \mathbf{u} - \tilde{\mathbf{u}}$ satisfies

$$\tilde{A}\mathbf{e} = h^2\mathbf{p} + h\mathbf{q}$$

We split $\mathbf{e} = \mathbf{e}_1 + \mathbf{e}_2$, so that

$$\tilde{A}\mathbf{e}_1 = h^2\mathbf{p} \quad \text{and} \quad \tilde{A}\mathbf{e}_2 = h\mathbf{q}$$

and use the norm property $\|\mathbf{e}\|_2 = \|\mathbf{e}_1 + \mathbf{e}_2\| \leq \|\mathbf{e}_1\|_2 + \|\mathbf{e}_2\|_2$

Global solution error (Mixed BC's)

Theorem: The global numerical solution error of the Poisson equation $-u'' = f$, $u(x_0) = 0$, $-u'(x_N) = \mu$ obtained with $\mathcal{O}(h^2)$ finite-difference discretization at x_i , $i = 1, \dots, N-1$; and $\mathcal{O}(h)$ discretization at x_N is $\|\mathbf{e}\|_{2,h} = \mathcal{O}(h^2)$

Proof: Exercise.

Hint: Find the eigenvalues of $\tilde{A}$ and show that $\|\mathbf{e}_1\|_{2,h} = \mathcal{O}(h^2)$. Show that $[\mathbf{e}_2]_i = ih^3q_N = h^2x_iq_N$ and use this fact to prove that $\|\mathbf{e}_2\|_{2,h} = \mathcal{O}(h^2)$.

Importance of knowing the discretization error

Why do we care about the order of discretization error?

- Analytical solutions of most PDE's are not known. Knowing $\mathcal{O}(h^p)$ will not tell you how big is the error for a given h , but will tell you how fast it will decrease if you refine your grid
- Suppose, the discretization error is $\mathcal{O}(h)$ and let $h_2 = h_1/k$. Then,

$$\frac{\|\mathbf{e}_{h_1}\|_{2,h}}{\|\mathbf{e}_{h_2}\|_{2,h}} = \frac{\mathcal{O}(h_1)}{\mathcal{O}(h_2)} \approx k$$

Whereas, with the discretization error of $\mathcal{O}(h^2)$

$$\frac{\|\mathbf{e}_{h_1}\|_{2,h}}{\|\mathbf{e}_{h_2}\|_{2,h}} = \frac{\mathcal{O}(h_1^2)}{\mathcal{O}(h_2^2)} \approx k^2$$

- Unfortunately, this behavior is only asymptotic, for $h \rightarrow 0$, and the global error is not always monotonically decreasing with h

$\mathcal{O}(h^2)$ is not everything: Convection-dominated problems

Convection-diffusion problem

Consider the steady-state convection-diffusion equation with Dirichlet BC's:

$$\begin{aligned} -\varepsilon c'' + vc' &= 0, \\ c(0) &= 0, \quad c(1) = 1 \end{aligned}$$

$c(x)$ – concentration of substance (e.g. pollutant)

ε – diffusion constant

v – velocity of the convecting medium

Analytical solution:

$$c(x) = \frac{e^{\frac{vx}{\varepsilon}} - 1}{e^{\frac{v}{\varepsilon}} - 1}$$

Péclet number, $Pe = vL/\varepsilon$, where L is the length of the domain, determines the relative "magnitude" of convection with respect to diffusion.

Standard $\mathcal{O}(h^2)$ FD approximation, inner points

$$\begin{aligned} -\varepsilon c_i'' + v c_i' &\approx \varepsilon \frac{-c_{i-1} + 2c_i - c_{i+1}}{h^2} + v \frac{c_{i+1} - c_{i-1}}{2h} \\ &= \varepsilon \left(\frac{-c_{i-1} + 2c_i - c_{i+1}}{h^2} + \frac{vh}{2\varepsilon} \frac{c_{i+1} - c_{i-1}}{h^2} \right) = 0 \end{aligned}$$

where mesh Péclet number is defined as

$$p_h = \frac{vh}{2\varepsilon} = \frac{h}{2L} \text{Pe} = \frac{1}{2N} \text{Pe}$$

FD equations

$$\begin{aligned} (-1 - p_h)c_{i-1} + 2c_i + (-1 + p_h)c_{i+1} &= 0 \\ i &= 2, 3, \dots, N-2 \end{aligned}$$

Standard $\mathcal{O}(h^2)$ FD approximation, boundary points

Left boundary

$$-\varepsilon c_1'' + v c_1' = 0, \text{ and } c_0 = 0$$

$$\frac{\varepsilon}{h^2} [(-1 - p_h)c_0 + 2c_1 + (-1 + p_h)c_2] = 0$$

$$2c_1 + (-1 + p_h)c_2 = 0$$

Right boundary

$$-\varepsilon c_{N-1}'' + v c_{N-1}' = 0, \text{ and } c_N = 1$$

$$\frac{\varepsilon}{h^2} [(-1 - p_h)c_{N-2} + 2c_{N-1} + (-1 + p_h)c_N]$$

$$= \frac{\varepsilon}{h^2} ((-1 - p_h)c_{N-2} + 2c_{N-1} + (-1 + p_h)1) = 0$$

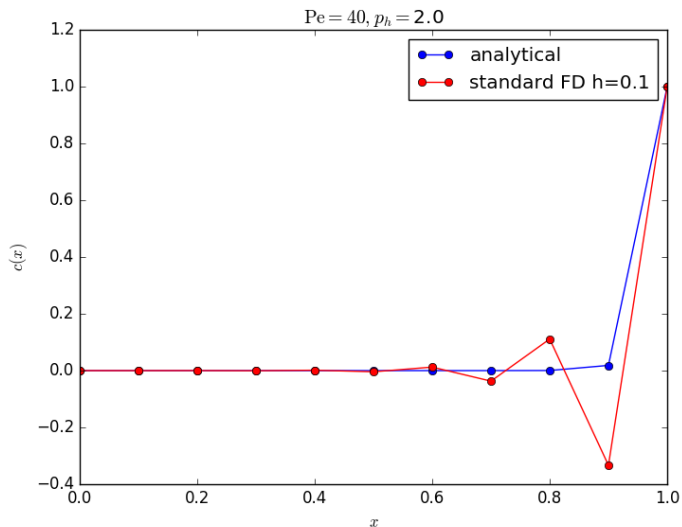
$$(-1 - p_h)c_{N-2} + 2c_{N-1} = 1 - p_h$$

Linear algebraic problem

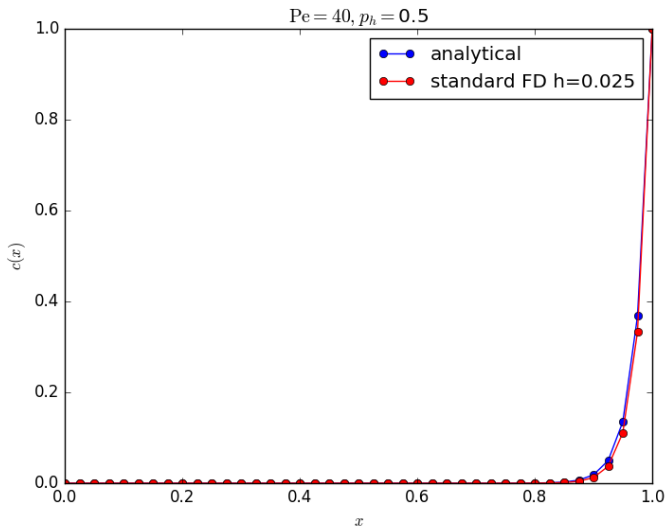
$$A\mathbf{c} = \mathbf{f}$$

$$A = \begin{bmatrix} 2 & -1 + p_h & 0 & 0 & \dots & 0 \\ -1 - p_h & 2 & -1 + p_h & 0 & \dots & 0 \\ \cdot & & & & & \cdot \\ \cdot & & & & & -1 + p_h \\ 0 & 0 & 0 & 0 & -1 - p_h & 2 \end{bmatrix}$$
$$\mathbf{f} = \begin{bmatrix} 0 \\ 0 \\ \cdot \\ \cdot \\ 1 - p_h \end{bmatrix}$$

Results with $h = 0.1$



Results with $h = 0.025$



Why?

Analytical solution of FD equation:

$$(-1 - p_h)c_{i-1} + 2c_i + (-1 + p_h)c_{i+1} = 0, \quad c_0 = 0, \quad c_N = 1$$

take: $c_i = r^i$

$$(-1 - p_h)r^{i-1} + 2r^i + (-1 + p_h)r^{i+1} = 0$$

$$-1 - p_h + 2r + (-1 + p_h)r^2 = 0$$

$$r = 1 \quad \text{and} \quad r = \frac{-1 - p_h}{-1 + p_h} = \frac{1 + p_h}{1 - p_h}$$

$$c_i = A + Br^i = A + B \left(\frac{1 + p_h}{1 - p_h} \right)^i$$

$$c_i = \frac{\left(\frac{1 + p_h}{1 - p_h} \right)^i - 1}{\left(\frac{1 + p_h}{1 - p_h} \right)^N - 1}, \quad \text{monotonically increasing if } \frac{1 + p_h}{1 - p_h} \geq 0$$

The mesh Péclet number must be $p_h = \nu h / (2\varepsilon) = \text{Pe } h / (2L) < 1$. Hence, the grid step must be $h < 2L / \text{Pe}$

The unwinding trick

The idea is to use the backward FD approximation $\mathcal{D}_x^-$, if $v > 0$, and the forward FD approximation $\mathcal{D}_x^+$, if $v < 0$. For example ($v > 0$):

$$\begin{aligned} -\varepsilon c_i'' + v c_i' &\approx \varepsilon \frac{-c_{i-1} + 2c_i - c_{i+1}}{h^2} + v \frac{c_i - c_{i-1}}{h} = 0 \\ (-1 - 2p_h)c_{i-1} + (2 + 2p_h)c_i - c_{i+1} &= 0 \end{aligned}$$

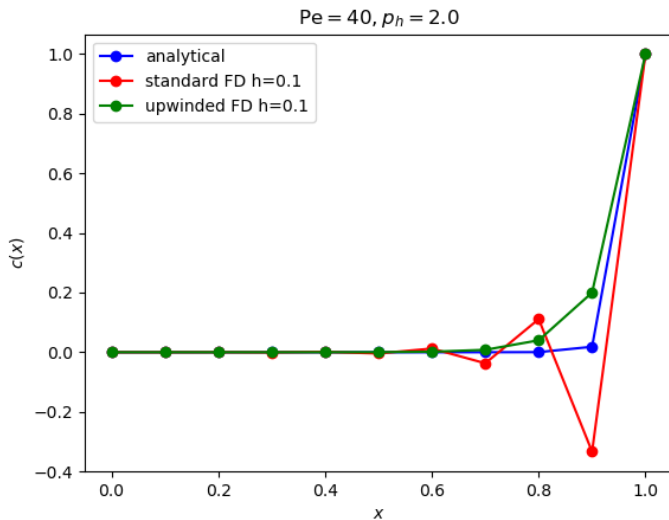
This FD equation has a monotonous solution for any $p_h > 0$. However, the approximation is $\mathcal{O}(h)$ now. Used mostly with shocks.

Linear algebraic problem

$$A\mathbf{c} = \mathbf{f}$$

$$A = \begin{bmatrix} 2 + 2p_h & -1 & 0 & 0 & \dots & 0 \\ -1 - 2p_h & 2 + 2p_h & -1 & 0 & \dots & 0 \\ \cdot & & & & & \cdot \\ \cdot & & & & & -1 \\ 0 & 0 & 0 & 0 & -1 - 2p_h & 2 + 2p_h \end{bmatrix}$$
$$\mathbf{f} = \begin{bmatrix} 0 \\ 0 \\ \cdot \\ \cdot \\ 1 \end{bmatrix}$$

Results with $h = 0.1$



Results with $h = 0.025$

